

# Curriculum Vitae

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## Education

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### MPI for Informatics & Saarland University

Saarbrücken, Germany

Ph.D. Student, Computer Science

Oct, 2022 - present

- Advisors: Prof. Bernt Schiele, Prof. Francesco Locatello

### International Institute of Information Technology, Hyderabad

Hyderabad, India

MS by Research in CSE

2018 - 2020

- Advisor: Prof. P. J. Narayanan
- Thesis: *Image Representations for Style Retrieval, Recognition and Background Replacement Tasks*
- CGPA: **9.75/10**

BTech with Honours in CSE

2014 - 2018

- CGPA: **8.91/10**

## Work Experience

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### Microsoft Research, India - Research Fellow

Aug 2020 - Aug 2022

- Advisors: Dr. Mohit Jain, Dr. Nipun Kwatra
- Developed a low-cost smartphone-based corneal topographer for keratoconus detection in collaboration with Sankara Eye Hospital. Published at ACM IMWUT/Ubicomp 2021.
- Worked on training DNN models for abnormality detection in limited data settings. Published at EMBC 2021.

### Microsoft Research, India - Research Intern

Jan 2020 - Jul 2020

- Advisors: Dr. Mohit Jain, Dr. Nipun Kwatra
- Worked on building smartphone based diagnostic solutions for eye diseases using mobile computing, computer vision and machine learning.

### Adobe Systems, India - Product Intern

Jun 2019 - Jan 2020

- Advisors: Balaji K., Mayur Hemani
- Worked at the Media and Data Science Research lab.
- Worked on Guided Few-shot Image Segmentation, improved on the prior state-of-the-art by 6%.
- Published at IJCAI 2020 and filed a patent.

### IIIT Hyderabad - Research Assistant

May 2016 - May 2019

- Advisor: Prof. P. J. Narayanan
- Research assistant at the Center for Visual Information and Technology (CVIT)
- Worked on representation learning for image style search and retrieval. Published at WACV 2020.
- Also worked on task of color-consistent background replacement. Published at MMM 2018.

### Google Summer of Code (GSoC) - Student Developer

May 2018 - Aug 2018

- Was accepted as a developer for the Google Summer of Code Program for the 2<sup>nd</sup> time with Scilab.
- Implemented a demo in C/C++ and Scilab as a working example for the MEX Library in Scilab.

### Google Summer of Code (GSoC) - Student Developer

May 2017 - Aug 2017

- Was accepted as a developer for the Google Summer of Code Program with the organization Scilab.
- Implemented a C/C++ wrapper for Matlab MEX-API on current API Scilab.

## Patents

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- **S. Gairola**, M. Hemani, A. Chopra, J. Dahl, Balaji K. Improved Similarity Propagation for One-Shot and Few-Shot Image Segmentation (US 16/906,954), 2020.

## Peer-Reviewed Publications

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- (10.) **S. Gairola**, M. Böhle, F. Locatello, B. Schiele. [How to Probe: Simple Yet Effective Techniques for Improving Post-hoc Explanations](#). In *International Conference on Learning Representations (ICLR)*, 2025.
- (9.) V. Ganatra, **S. Gairola**, P. Joshi, A. Balasubramaniam, K. Murali, A. Varadharajan, B. Mallikarjuna, N. Kwatra, M. Jain. [SmartKC++: Improving Performance of Smartphone-Based Corneal Topographers](#). In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2025.
- (8.) **S. Gairola**, P. Joshi, A. Balasubramaniam, K. Murali, N. Kwatra, M. Jain. [Keratoconus Classifier for Smartphone-based Corneal Topographer](#). *44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)* (2022).
- (7.) A. Aggarwal, **S. Gairola**, U. Uppadhyay, A. P. Vasishta, D. Rao, A. Goyal, K. Murali, N. Kwatra, and M. Jain. [Towards Automating Retinoscopy for Refractive Error Diagnosis](#). In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, Volume 6, Issue 3, 2022.
- (6.) **S. Gairola**, M. Bohra, N. Shaheer, N. Jayaprakash, P. Joshi, A. Balasubramaniam, K. Murali, N. Kwatra, and M. Jain. [SmartKC: Smartphone-based Corneal Topographer for Keratoconus Detection](#). In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, Volume 5, Issue 4, 2021.
- (5.) **S. Gairola**, F. Tom, N. Kwatra, and M. Jain. [RespireNet: A Deep Neural Network for Accurately Detecting Abnormal Lung Sounds in Limited Data Setting](#). *43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2021.
- (4.) **S. Gairola**, A. Chopra, M. Hemani, Balaji K. [SimPropNet: Improved Similarity Propagation for Few-Shot Image Segmentation](#). *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.
- (3.) **S. Gairola**, R. Shah, P. J. Narayanan. [Unsupervised Image Style Embeddings for Retrieval and Recognition Tasks](#). *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2020.
- (2.) V. Kumar\*, D. Khattar\*, **S. Gairola\***, Y. K. Lal\*, V. Varma. [Identifying Clickbait: A Multi-Strategy Approach Using Neural Networks](#). *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, 2018.
- (1.) S. Rawat\*, **S. Gairola\***, R. Shah, P. J. Narayanan. [Find Me a Sky: A Data-Driven Method for Color-Consistent Sky Search and Replacement](#). *International Conference on Multimedia Modeling (MMM)*, 2018.

\*equal contribution

## Academic Services

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**Organizer:** [Computer Vision for Developing Countries Workshop at ACCV 2024](#)

**Reviewer:** ICML 2023-25; NeurIPS 2022-23; ICLR 2022-24; CVPR 2024-25; ECCV 2024; ICCV 2025; IHCI 2021

**Volunteer:** ICVGIP 2018

## Talks & Conference Presentations

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Intriguing Applications and Overlooked Pitfalls of XAI in Visual Models, at the [GMUM Workshop, Jagiellonian University](#), October 2024.

A DNN for Accurately Detecting Abnormal Lung Sounds in Limited Data Setting, *EMBC 2021* ([video](#)).

Improved Similarity Propagation for Few-shot Image Segmentation, *IJCAI 2020* ([video](#)).

Unsupervised Image Style Embeddings for Retrieval and Recognition Tasks, *WACV 2020* ([video](#)).

## Selected Research Projects

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### Smartphone-based Corneal Topographer

Nov 2020 - Oct 2021

Developed a low-cost smartphone-based corneal topographer. It provides a portable and scalable solution for the mass screening of keratoconus. Built the entire system: an AI powered smartphone app, the 3D printed attachment, and the processing software. [*Android, Python, PyTorch*; [project page](#)]

### Abnormality Detection in Respiratory Signals

Aug 2020 - Nov 2020

Developed a ‘ResNet34’ based model along with a suite of novel augmentation and fine-tuning techniques to effectively utilize small-sized datasets. The method improved upon the state-of-the-art for 4-class classification on the ICBHI dataset by 2.2%. [*Python, PyTorch*; [code](#)]

### Similarity Propagation for Few-Shot Image Segmentation

Jun 2019 - Jan 2020

Developed a deep learning system that leverages background-foreground similarity to perform few-shot image segmentation. The method achieved state-of-the-art results on the PASCAL-5i, COCO-20i and FSS datasets. [*Python, PyTorch*]

### Unsupervised Image Style Representation Learning

Jan 2019 - Jun 2019

Developed a deep learning model to learn robust style representations without supervision. Achieved state-of-the-art results across six datasets and it was published as a paper. [*Python, PyTorch*; [code](#)]

### Sketching with Style

Aug 2018 - Dec 2018

Implemented the ICCV 2017 paper “Sketching with Style” ([link](#)). Was able to reproduce the results provided in the paper and open-sourced the code. [*PyTorch, Python*; [code](#)]

### Neural Clickbait Detection Engine

May 2017 - Aug 2017

Developed a system to robustly identify clickbait social media posts. The system comprises of a siamese-network that fuses visual and textual information to learn a neural embedding used to detect clickbaits. [*Python, Keras*; [code](#)]

### Color Consistent Background Replacement

Jan 2017 - Apr 2017

Developed a data driven method for color-consistent sky search and replacement. A diverse set of skies consistent in color and illumination are retrieved from a curated dataset, and used to generate composites. The composites are re-ranked based on realism and diversity. [*MATLAB*; [project page](#)]

## Awards & Achievements

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- Awarded the **Dean’s Research Award** for a paper published at an international conference while in undergraduate, 2018.
- Awarded the **Dean’s List Award** for 6 semesters straight: Monsoon’15, ’16, ’17 & Spring’16, ’17, ’18
- Qualified for the **ACM ICPC Asia Onsite Regionals** twice (2015, 2016).
- **98.5** percentile in **JEE Advanced (2014)** examination among 150,000 students.
- **99.7** percentile in **JEE Main (2014)** examination among 1.2 million students.

## Teaching Experience

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**Saarland University:** Elements of Data Science and Artificial Intelligence (Winter 2023, 2024)

**IIIT Hyderabad:** Computer Graphics (Spring 2019), Digital Image Processing (Monsoon 2017, 2018), Computer Vision (Spring 2018), Artificial Intelligence (Spring 2017), Digital Logic and Processors (Monsoon 2016)

The duties as a TA involved taking regular tutorials, setting up assignments and conducting evaluations.

## Technical Skills

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|---------------------------------|----------------------------------------------------------|
| <b>Working Knowledge</b>        | Bash, C/C++, Python, MATLAB, Octave, OpenCV, Android-SDK |
| <b>Software &amp; Tools</b>     | GNU/Linux, HTML, CSS, JavaScript, LaTeX, Excel           |
| <b>Deep Learning Frameworks</b> | PyTorch, TensorFlow, Keras, Caffe                        |
| <b>Past Experience</b>          | Java, OpenGL, WebGL, Django, Web2py                      |